Chi-Square Tests & Regression Inference

ANSWER KEY

Section 1: Chi-Square Goodness-of-Fit Test

1 What is the formula for the Chi-Square test statistic?

a. $\chi^2 = \sum \frac{(O-E)^2}{E}$ ✓

b. $\chi^2 = \sum \frac{(O-E)}{E}$

c. $\chi^2 = \sum \frac{(O-E)^2}{O}$

d. $\chi^2 = \frac{(\bar{x}-\mu)^2}{s^2}$

The Chi-square statistic measures discrepancy between observed counts O and expected counts E: $\chi^2 = \sum \frac{(O-E)^2}{E}$.

2 A bag of candies claims to have 5 colors in equal proportions (20% each). In a sample of 100 candies, what are the degrees of freedom for the goodness-of-fit test?

a. 4 ✓

b. 5

c. 99

d. 100

For a goodness-of-fit test with k categories, degrees of freedom is $df = k - 1$. Here $df = 5 - 1 = 4$.

3 What condition must be satisfied regarding expected counts for a Chi-Square test to be valid?

a. **All expected cell counts must be at least** 5 ✓

b. All observed cell counts must be at least 30

c. The sample size must be exactly 100

d. The categories must be normally distributed

The standard Large Counts condition for Chi-square tests requires all expected counts to be at least 5.

Section 2: Chi-Square Test of Independence vs. Homogeneity

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4 In a two-way contingency table with r rows and c columns, how is the expected count for each cell calculated under H_0?

a. $E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Table Total}}$ ✓

b. $E = \frac{\text{Row Total} + \text{Column Total}}{2}$

c. $E = \frac{\text{Table Total}}{r \times c}$

d. $E = \frac{\text{Row Total}}{\text{Column Total}}$

Under independence, $P(\text{Row} \cap \text{Col}) = P(\text{Row})P(\text{Col})$, giving $E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$.

5 A single random sample of 300 voters is categorized by political affiliation and stance on a ballot issue. Which test assesses if affiliation and stance are associated?

a. **Chi-Square Test of Independence** ✓

b. Chi-Square Test of Homogeneity

c. Chi-Square Goodness-of-Fit Test

d. Two-sample z-test for proportions

When a single random sample is classified on two categorical variables, the test of association is the Chi-Square Test of Independence.

6 What are the degrees of freedom for a Chi-Square test on a 3×4 contingency table?

a. 6 ✓

b. 12

c. 7

d. 11

$df = (r-1)(c-1) = (3-1)(4-1) = 2 \times 3 = 6.$

Section 3: Inference for Linear Regression Slope

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7 In testing for a significant linear relationship between x and y, what are the standard null and alternative hypotheses for the population slope β?

a. $H_0 : \beta = 0$ **versus** $H_a : \beta \neq 0$ ✓

b. $H_0 : \beta = 1$ versus $H_a : \beta \neq 1$

c. $H_0 : b = 0$ versus $H_a : b \neq 0$

d. $H_0 : r = 1$ versus $H_a : r \neq 1$

The null hypothesis states there is no linear relationship (population slope $\beta = 0$), tested against $\beta \neq 0$ (or a directional alternative).

8 Computer regression output shows: Predictor Coef SE Coef T P; Constant 14.2 2.1 6.76 0.000; Slope 3.45 0.69 5.00 0.0001. For $n = 18$ observations, what is the t-statistic for the slope?

a. 5.00 ✓

b. 6.76

c. 3.45

d. 0.69

The t-statistic for the slope is $t = \frac{b - 0}{\text{SE}_b} = \frac{3.45}{0.69} = 5.00$.

9 What are the degrees of freedom for the t-test of the slope in simple linear regression with n data pairs?

a. $n - 2$ ✓

b. $n - 1$

c. n

d. $2n - 2$

In simple linear regression, two parameters are estimated (intercept α and slope β), leaving $df = n - 2$ degrees of freedom.

Section 4: Regression Model Conditions (LINER)

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10 What does the acronym LINER stand for regarding the conditions for regression inference?

a. **Linear, Independent, Normal y for each x, Equal variance, Random sampling ✓**

b. Least squares, Intercept, Normal, Errors, Residuals

c. Linear, Integrated, Numerical, Estimated, Robust

d. Limit, Interval, Normal, Expected, Ratio

The regression conditions are: Linear relationship, Independent observations, Normal distribution of response y at each x, Equal variance of residuals (homoscedasticity), and Random data collection.

11 If a residual plot displays a funnel or cone shape (spread widens as x increases), which condition is violated?

a. **Equal variance of residuals (homoscedasticity) ✓**

b. Linearity

c. Independence

d. Random sampling

A fan or funnel shape indicates heteroscedasticity (non-constant variance), violating the equal variance condition.

12 A 95% confidence interval for the regression slope β is $(1.24, 3.86)$. What can be concluded at $\alpha = 0.05$?

a. **Because 0 is not in the interval, there is convincing evidence of a positive linear relationship between x and y ✓**

b. Because 0 is not in the interval, the relationship is non-linear

c. The correlation coefficient r must equal 1.24

d. We fail to reject $H_0 : \beta = 0$

Since the entire confidence interval is strictly positive and excludes 0, we reject $\beta = 0$ at the 5% level, confirming a statistically significant positive linear relationship.

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